CSA0501

DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENT

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EXP 9:

AIM

To create view and index on the given relation.

ALGORITHM

1. Start.
2. Create a database and select it.
3. Create the EMPLOYEES1 table with appropriate columns.
4. Insert sample records into the table.
5. Create a VIEW (v1) on selected columns of EMPLOYEES1 using CREATE VIEW ... AS SELECT.
6. Perform INSERT and DELETE operations through the view and observe the effect on the base table.
7. Create an INDEX on the salary attribute, retrieve employees with salary above a threshold, and use EXPLAIN to observe indexing behavior; then drop the index.
8. Create an INDEX on the employee_id attribute, retrieve a specific employee's details, and use EXPLAIN to observe indexing behavior.
9. Display and verify the output of each operation.
10. Stop.

COMMANDS (short format)

sql

```
CREATE DATABASE dbms_lab10;

USE dbms_lab10;

CREATE TABLE employees1 (...);

INSERT INTO employees1 VALUES (...);

CREATE VIEW v1 AS SELECT employee_id, salary FROM employees1;
```

INSERT INTO v1 VALUES (101, 100001);

DELETE FROM v1 WHERE employee_id = 10;

CREATE INDEX index1 ON employees1(salary);

SELECT first_name FROM employees1 WHERE salary > 145000;

EXPLAIN SELECT first_name FROM employees1 WHERE salary > 145000;

DROP INDEX index1 ON employees1;

CREATE INDEX index1 ON employees1(employee_id);

SELECT first_name FROM employees1 WHERE employee_id = 10;

EXPLAIN SELECT first_name FROM employees1 WHERE employee_id = 10;

QUESTIONS:

On VIEW

1. Create a view with name 'v1' using employees1 table which holds the value of employee_id and salary of employee.
2. Do the insert and delete records from v1 table.

```
mysql> SELECT * FROM employees1;
```

employee_id	first_name	last_name	device_serial	salary
1	Arun	Kumar	ABC123	60000
2	Divya	Raj	DEF456	65000
3	Bala	Murugan	GHI789	70000
4	Shalini	Reddy	JKL012	75000
5	Manoj	Iyer	MNO345	80000
6	Ezhil	Prakash	PQR678	85000
7	Deepak	William	STU901	90000
8	Sarah	Johnson	VWX234	95000
9	Jagan	Brown	YZA567	100000
10	Meena	Miller	BCD890	105000
11	Vishal	Davis	EFG123	110000
12	Oviya	Garcia	HIJ456	115000
13	Chris	Rodriguez	KLM789	120000
14	Isha	Wilson	NOP012	125000
15	Mathew	Martinez	QRS345	130000
16	Sophia	Anderson	TUV678	135000
17	Daniel	Smith	WXY901	140000
18	Mia	Thomas	ZAB234	145000
19	Joseph	Hernandez	CDE567	150000
20	Abigail	Smith	FGH890	155000

```
20 rows in set (0.01 sec)
```

```
mysql> DESC employees1;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| employee_id   | int         | NO   | PRI | NULL    |       |
| first_name    | varchar(50) | YES  |     | NULL    |       |
| last_name     | varchar(50) | YES  |     | NULL    |       |
| device_serial | varchar(15) | YES  |     | NULL    |       |
| salary        | int         | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
5 rows in set (0.12 sec)

mysql> CREATE VIEW v1 AS
    -> SELECT employee_id, salary FROM employees1;
Query OK, 0 rows affected (0.03 sec)

mysql> DESC v1;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| employee_id | int  | NO   |     | NULL    |       |
| salary      | int  | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
2 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM v1;
+-------------+--------+
| employee_id | salary |
+-------------+--------+
|           1 |  60000 |
|           2 |  65000 |
|           3 |  70000 |
|           4 |  75000 |
|           5 |  80000 |
|           6 |  85000 |
|           7 |  90000 |
|           8 |  95000 |
|           9 | 100000 |
|          10 | 105000 |
|          11 | 110000 |
|          12 | 115000 |
|          13 | 120000 |
|          14 | 125000 |
|          15 | 130000 |
|          16 | 135000 |
|          17 | 140000 |
|          18 | 145000 |
|          19 | 150000 |
|          20 | 155000 |
+-------------+--------+
20 rows in set (0.01 sec)
```

```
mysql>
mysql> DELETE FROM v1 WHERE employee_id = 10;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM v1;
+-------------+--------+
| employee_id | salary |
+-------------+--------+
|           1 |  60000 |
|           2 |  65000 |
|           3 |  70000 |
|           4 |  75000 |
|           5 |  80000 |
|           6 |  85000 |
|           7 |  90000 |
|           8 |  95000 |
|           9 | 100000 |
|          11 | 110000 |
|          12 | 115000 |
|          13 | 120000 |
|          14 | 125000 |
|          15 | 130000 |
|          16 | 135000 |
|          17 | 140000 |
|          18 | 145000 |
|          19 | 150000 |
|          20 | 155000 |
|         101 | 100001 |
+-------------+--------+
20 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM employees1;
+-------------+------------+-----------+---------------+--------+
| employee_id | first_name | last_name | device_serial | salary |
+-------------+------------+-----------+---------------+--------+
|           1 | Arun       | Kumar     | ABC123        |  60000 |
|           2 | Divya      | Raj       | DEF456        |  65000 |
|           3 | Bala       | Murugan   | GHI789        |  70000 |
|           4 | Shalini    | Reddy     | JKL012        |  75000 |
|           5 | Manoj      | Iyer      | MNO345        |  80000 |
|           6 | Ezhil      | Prakash   | PQR678        |  85000 |
|           7 | Deepak     | William   | STU901        |  90000 |
|           8 | Sarah      | Johnson   | VWX234        |  95000 |
|           9 | Jagan      | Brown     | YZA567        | 100000 |
|          11 | Vishal     | Davis     | EFG123        | 110000 |
|          12 | Oviya      | Garcia    | HIJ456        | 115000 |
|          13 | Chris      | Rodriguez | KLM789        | 120000 |
|          14 | Isha       | Wilson    | NOP012        | 125000 |
|          15 | Mathew     | Martinez  | QRS345        | 130000 |
|          16 | Sophia     | Anderson  | TUV678        | 135000 |
|          17 | Daniel     | Smith     | WXY901        | 140000 |
|          18 | Mia        | Thomas    | ZAB234        | 145000 |
|          19 | Joseph     | Hernandez | CDE567        | 150000 |
|          20 | Abigail    | Smith     | FGH890        | 155000 |
|         101 | NULL       | NULL      | NULL          | 100001 |
+-------------+------------+-----------+---------------+--------+
20 rows in set (0.00 sec)
```

On INDEX

1. Create index1 for 'salary' attribute from employees1 relation and list the first name of the employees whose salary is above 145000 and explain the working principle of indexing and then drop index1.
2. Create index1 for 'employee_id' attribute and display the first name of an employee whose employee id is 10 and explain the working principle of index1.

```
mysql> CREATE INDEX index1 ON employees1(salary);
Query OK, 0 rows affected (0.21 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT first_name FROM employees1 WHERE salary > 145000;
+------------+
| first_name |
+------------+
| Joseph     |
| Abigail    |
+------------+
2 rows in set (0.01 sec)
```

```
mysql> EXPLAIN SELECT first_name FROM employees1 WHERE salary > 145000;
+----+-------------+------------+------------+-------+---------------+--------+---------+------+------+----------+------
-----------------+
| id | select_type | table      | partitions | type  | possible_keys | key    | key_len | ref  | rows | filtered | Extra
                 |
+----+-------------+------------+------------+-------+---------------+--------+---------+------+------+----------+------
-----------------+
|  1 | SIMPLE      | employees1 | NULL       | range | index1        | index1 | 5       | NULL |    2 |   100.00 | Using
 index condition |
+----+-------------+------------+------------+-------+---------------+--------+---------+------+------+----------+------
-----------------+
1 row in set, 1 warning (0.01 sec)
```

```
mysql> DROP INDEX index1 ON employees1;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX index1 ON employees1(employee_id);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT first_name FROM employees1 WHERE employee_id = 10;
Empty set (0.00 sec)

mysql> EXPLAIN SELECT first_name FROM employees1 WHERE employee_id = 10;
+----+-------------+-------+------------+------+---------------+------+---------+------+------+----------+--------------
------------------+
| id | select_type | table | partitions | type | possible_keys | key  | key_len | ref  | rows | filtered | Extra
                  |
+----+-------------+-------+------------+------+---------------+------+---------+------+------+----------+--------------
------------------+
|  1 | SIMPLE      | NULL  | NULL       | NULL | NULL          | NULL | NULL    | NULL | NULL |     NULL | no matching r
ow in const table |
+----+-------------+-------+------------+------+---------------+------+---------+------+------+----------+--------------
------------------+
1 row in set, 1 warning (0.00 sec)
```

```
mysql> SHOW INDEX FROM employees1;
+------------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+
------------+---------+---------------+---------+------------+
| Table      | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null |
 Index_type | Comment | Index_comment | Visible | Expression |
+------------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+
------------+---------+---------------+---------+------------+
| employees1 |          0 | PRIMARY  |            1 | employee_id | A         |          20 |     NULL |   NULL |      |
 BTREE      |         |               | YES     | NULL       |
| employees1 |          1 | index1   |            1 | employee_id | A         |          20 |     NULL |   NULL |      |
 BTREE      |         |               | YES     | NULL       |
+------------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+
------------+---------+---------------+---------+------------+
2 rows in set (0.07 sec)
```

RESULT

The records from the tables are displayed using View and Index on the given relation.

Ready for the next experiment whenever you have the manual pages.